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MULTIVARIABLE CALCULUS • UNIT 4.01

Double Integrals, Polar Coordinates & Jacobians

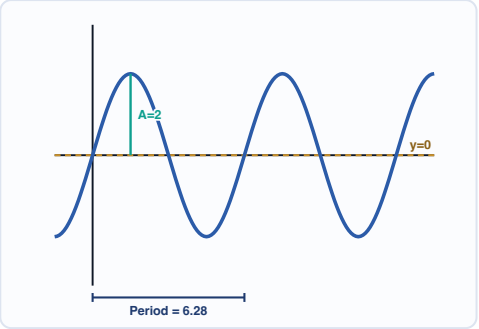
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Double Integrals, Polar Coordinates & Jacobians Practice Exercises

1 MCQ

Compute $\iint_R dA$ for $R = [0, 4] \times [0, 3]$ (the area of R).



A 12

B 7

C 14

D 6

2 MCQ

Compute $\iint_R dA$ for $R = [0, 2] \times [0, 4]$ (the area of R).

A 8

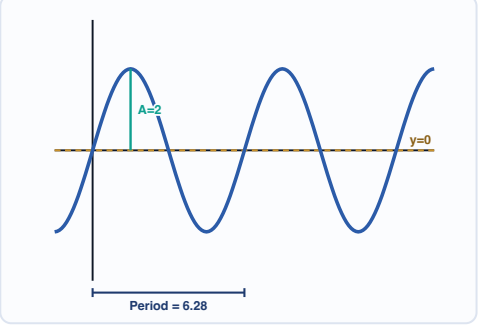
B 6

C 12

D 4

3 MCQ

Compute $\iint_R dA$ for $R = [0, 4] \times [0, 5]$ (the area of R).



A 20

B 9

C 18

D 10

4 MCQ

Compute $\iint_R dA$ for $R = [0, 2] \times [0, 8]$ (the area of R).

A 10

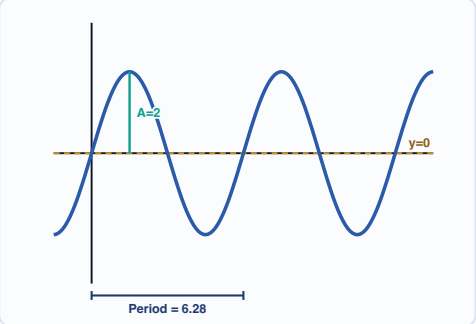
B 16

C 20

D 8

5 MCQ

Compute $\iint_R dA$ for $R = [0, 7] \times [0, 8]$ (the area of R).



A 15

B 30

C 56

D 28

6 MCQ

Compute $\iint_R dA$ for $R = [0, 8] \times [0, 3]$ (the area of R).

A 11

B 22

C 12

D 24

7 MCQ

Compute $\iint_R dA$ for $R = [0, 1] \times [0, 3]$ (the area of R).

A 4

B 8

C 3

D 1

8 MCQ

Compute $\iint_R dA$ for $R = [0, 5] \times [0, 3]$ (the area of R).

A 8

B 16

C 15

D 13

9 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	2

A 18

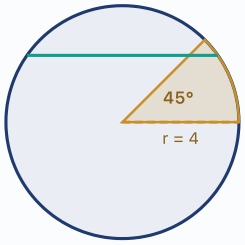
B 12

C 6

D 24

10 MCQ

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2+y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.



A $\pi/4$

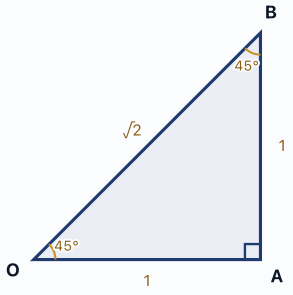
B 4

C 16

D 2

11 MCQ

Evaluate the double integral $\iint_R xy\, dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.



A 1/12

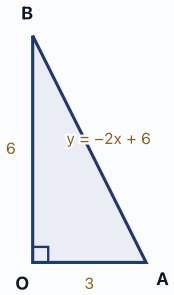
B 1/6

C 1/8

D 1/4

12 MCQ

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\, dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\, dr\, d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.



A $\alpha = 0, \beta = \frac{\pi}{4}$

B $\alpha = 0, \beta = \frac{3\pi}{4}$

C $\alpha = 0, \beta = \frac{\pi}{2}$

D $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

13 MCQ

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\, dA$.

A $\frac{1}{12}$

B $\frac{1}{8}$

C $\frac{1}{6}$

D $\frac{1}{10}$

14 MCQ

Find $\nabla f(-5, -5)$ for $f(x, y) = 9x^2 + 6y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	9	6	15

A (90, 60)

B (-60, -90)

C (-90, -60)

D (-45, -30)

15 MCQ

Deduce $\nabla f(-2, 5)$ for $f(x, y) = 2x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	8	10

A (8, -80)

B (80, -8)

C (-8, 80)

D (-4, 40)

16 MCQ

Obtain $\nabla f(-1, 3)$ for $f(x, y) = 8x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	9	17

A (16, -54)

B (54, -16)

C (-16, 54)

D (-8, 27)

17 MCQ

Identify $\nabla f(-2, 3)$ for $f(x, y) = 2x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	8	10

A (8, -48)

B (48, -8)

C (-4, 24)

D (-8, 48)

18 MCQ

Identify $\nabla f(-3, -2)$ for $f(x, y) = 9x^2 + 4y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	9	4	13

A (-16, -54)

B (-54, -16)

C (-27, -8)

D (54, 16)

19 MCQ

Compute $\nabla f(4, 6)$ for $f(x, y) = 7x^2 + 5y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	7	5	12

A (-56, -60)

B (56, 60)

C (60, 56)

D (28, 30)

20 MCQ

Find $\nabla f(-1, -2)$ for $f(x, y) = 5x^2y + 3xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	8

A (20, 12)

B (17, 32)

C (-32, -17)

D (32, 17)

21 MCQ

Find $\nabla f(-4, 3)$ for $f(x, y) = 5x^2y + xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	6

A (56, -111)

B (111, -56)

C (-111, 56)

D (-120, -24)

22 MCQ

Calculate $\nabla f(1, 1)$ for $f(x, y) = 5x^2y + 2xy^2$.

A (12, 9)

B (9, 12)

C (-12, -9)

D (10, 4)

23 MCQ

Find $\nabla f(1, -1)$ for $f(x, y) = 4x^2y + xy^2$.

A (-7, 2)

B (-8, -2)

C (2, -7)

D (7, -2)

24 MCQ

Obtain $\nabla f(-1, 4)$ for $f(x, y) = 5x^2y + 4xy^2$.

A (-27, 24)

B (-24, 27)

C (-40, -32)

D (24, -27)

25 MCQ

Compute $\nabla f(-1, -1)$ for $f(x, y) = 4x^2y + 2xy^2$.

A (8, 4)

B (8, 10)

C (10, 8)

D (-10, -8)